

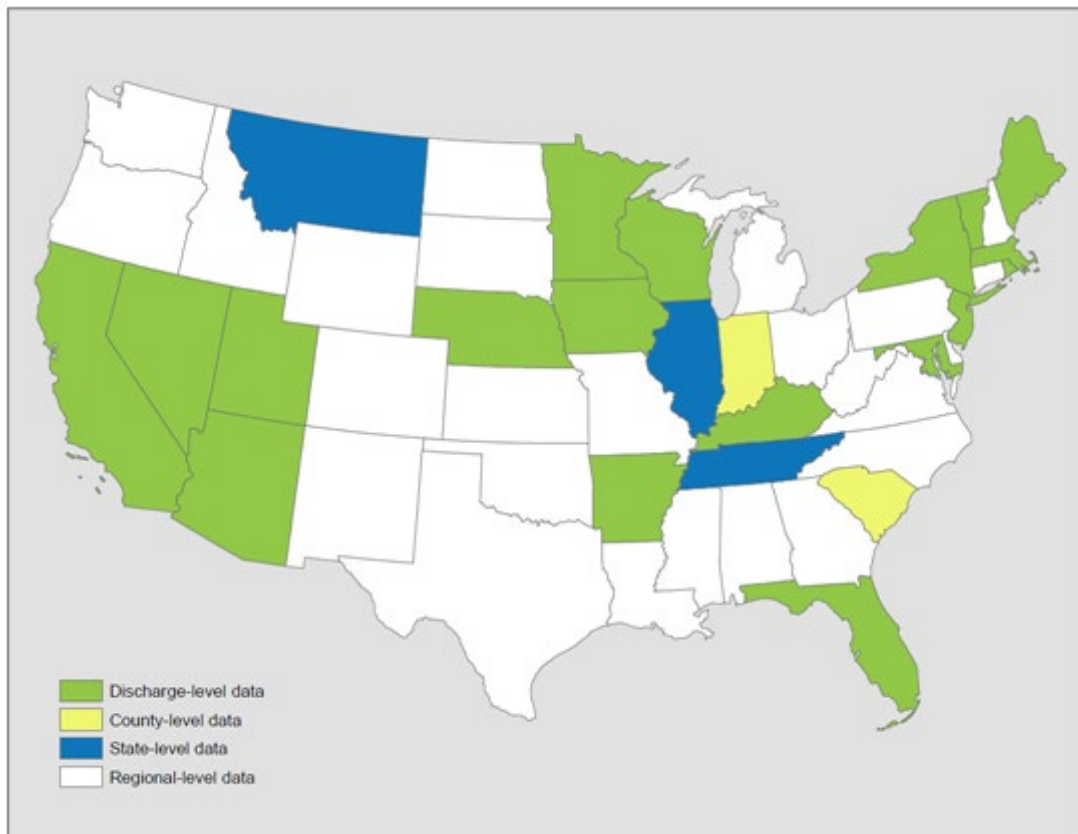
785,000 new heart attacks per year (Roger et al., 2012). Exclusion of heart attack deaths reported by CDC Wonder yields approximately 575,000 nonfatal cases per year.

An alternative approach to the estimation of heart attack rates is to use data from the Healthcare Cost and Utilization Project (HCUP), assuming that all heart attacks that are not instantly fatal will result in a hospitalization. Details about HCUP data are described in Section D.2. According to the 2014 HCUP data there were approximately 608,795 hospitalizations due to heart attacks (acute myocardial infarction: ICD-9 410, primary diagnosis). We used rates based on HCUP data over estimates extrapolated from cohort studies because the former is a national database with a larger sample size, which is intended to provide reliable national estimates. The incidence rate calculation is also described in Section D.2 and the incidence rates for AMI hospitalization are presented in Table D-10.

Rosamond et al. (1999) reported that approximately six percent of male and eight percent of female hospitalized heart attack patients die within 28 days (either in or outside of the hospital). We, therefore, applied a factor of 0.93 to the estimated number of PM-related acute myocardial infarctions to exclude the number of cases that result in death within the first month. Note that we did not adjust for fatal AMIs in the incidence rate estimation, due to the way that the epidemiological studies are designed. Those studies consider total admissions for AMIs, which includes individuals living at the time the studies were conducted. Therefore, we use the definition of AMI that matches the definition in the epidemiological studies.

D.4 Emergency Department Visits

The data source for emergency department/room (ED or ER) visits is also HCUP, i.e., SID, SEDD, and NEDS. And the types of data providers are also the same as those described in Section D.2. Figure D-2 shows the emergency department data in each state.

Figure D-2. Emergency Department Data from HCUP

The calculation of ER visit rates is also similar to the calculation of hospitalization rates, except for the following differences:

- The SEDD databases include only those ER visits that ended with discharge. To identify the ER visits that ended in hospitalization, we used a variable called “admission source” in the SID databases. Admission source identified as “emergency room” indicates that the hospital admission came from the ER - i.e., the ER visit ended in hospitalization. For each combination of age group, endpoint and county, we summed the ER visits that ended with discharge and those that resulted in hospitalization.
- The data year varies across the states from 2011 to 2014; we assumed that ER visit rates are reasonably constant across these three years and consider them as 2014 rates.
- Instead of using HCUPnet/NIS in the last step as described in Section D.2., we used HCUPnet/NEDS to calculate ER visit rates for states without discharge level or state level data. Table D-11 presents the estimated asthma emergency room rates by health endpoint and age group.

Table D-11. Emergency Department Visit Rates (per 100 people per year) by Health Endpoint and Age Group

Emergency Department Category	ICD-9 Codes	Age 0-17	18-24	25-34	35-44	45-54	55-64	65-74	75-84	85+
Asthma	493	0.959	0.601	0.556	0.538	0.552	0.408	0.331	0.368	0.350
Respiratory	491-493, 460-466, 477.0-477.9, 480-486, 496, 786.07, 786.09	6.069	3.214	2.837	2.332	2.447	2.418	2.908	4.382	5.651
Cardiovascular	410-414, 427-428, 433-437, 440.0-440.9, 443-445, 451-453	0.030	0.107	0.212	0.496	1.151	2.023	3.451	6.726	11.028
All Cardiac Outcomes	390-459	0.067	0.314	0.568	1.105	2.021	3.086	4.921	9.345	14.596

D.5 School Loss Days

Epidemiological studies have examined the relationship between air pollution and a variety of measures of school absence. These measures include: school loss days for all causes, illness-related, and respiratory illness-related. We have two sources of information. The first is the National Center for Education Statistics, which provided an estimate of all-cause school loss days, and the other is the National Health Interview Survey (Adams et al., 1999, Table 47), which has data on different categories of acute school loss days. Table D-12 presents the estimated school loss day rates. Further detail is provided below on these rates.

Table D-12. School Loss Day Rates (per student per year)

Type	Northeast	Midwest	South	West
Respiratory illness-related absences	1.3	1.7	1.1	2.2
Illness-related absences	2.4	2.6	2.6	3.7
All-cause	9.9	9.9	9.9	9.9

* We based illness-related school loss day rates on data from the 1996 NHIS and an estimate of 180 school days per year. This excludes school loss days due to injuries. We based the all-cause school loss day rate on data from the National Center for Education Statistics.

All-Cause School Loss Day Rate

Based on data from the U.S. Department of Education (1996, Table 42-1), the National Center for Education Statistics estimates that for the 1993-1994 school year, 5.5 percent of students are absent from school on a given day. This estimate is comparable to study-specific estimates from Chen et al. (2000) and Ransom and Pope (1992), which ranged from 4.5 to 5.1 percent.

Illness-Related School Loss Day Rate

The National Health Interview Survey (NHIS) has regional estimates of school loss days due to a variety of acute conditions (Adams et al., 1999). NHIS is a nationwide sample-based survey of the health of the noninstitutionalized, civilian population, conducted by NCHS. The survey collects data on acute conditions, prevalence of chronic conditions, episodes of injury, activity limitations, and self-reported health status. However, it does not provide an estimate of all-cause school loss days.

In estimating illness-related school loss days, we started with school loss days due to acute problems (Adams et al., 1999, Table 47) and subtracted lost days due to injuries, in order to match the definition of the study used in the C-R function to estimate illness-related school absences (Gilliland et al., 2001). We then divided by 180 school days per to estimate illness-related school absence rates per school day. Similarly, when estimating respiratory illness-related school loss days, we use data from Adams et al. (1999, Table 47). Note that we estimated 180 school days in a year to calculate respiratory illness-related school absence rates per year.

D.6 Other Acute and Chronic Effects

For many of the minor effect studies, baseline rates from a single study are often the only source of information, and we assume that these rates hold for locations in the U.S. The use of study-specific estimates are likely to increase the uncertainty around the estimate because they are often estimated from a single location using a relatively small sample. These endpoints include: acute bronchitis, chronic bronchitis, upper respiratory symptoms, lower respiratory symptoms. Table D-13 presents a summary of these baseline rates.

Table D-13. Selected Acute and Chronic Incidence (Cases / Person-Year) & Prevalence (Percentage Population)

Endpoint	Age	Parameter	Rate	Source
Acute Bronchitis	8-12	Incidence	0.043	American Lung Association (2002b, Table 11)
Chronic Bronchitis	27+	Incidence	0.00378	Abbey et al. (1993, Table 3)
Chronic Bronchitis	18+	Prevalence	4.37%	American Lung Association (2010a, Table 4). The rate numbers may be slightly different from those in Table 4 because we received more current estimates form ALA.
	18-44		3.15%	
	45-64		5.49%	
	65+		5.63%	
Lower Respiratory Symptoms (LRS)	7-14	Incidence	0.483	Schwartz et al. (1994, Table 2)
Minor Restricted Activity Days (MRAD)	18-64	Incidence	7.8	Ostro and Rothschild (1989, p. 243)
Work Loss Day (WLD)	18-64	Incidence	2.172	Adams et al. (1999, Table) U.S.